

99 Per Cent of the Procedure, 27 Per Cent of the Code:  
The Enacted Law and the Applied Law in 50,666 Judgments

Abdullah Almohammed

Independent Researcher

abdullah.m.almohammed@gmail.com

ORCID: 0009-0001-0832-0995

Correspondence, proofs and offprints:

Abdullah Almohammed

[postal address withheld from the public repository]

Telephone: [telephone withheld]

abdullah.m.almohammed@gmail.com

## Abstract

A legislature enacts a statute book; courts apply some part of it. How large that part is has rarely been measured, because it requires matching what judgments cite to the articles that exist, at the level of the article. This article does so for one jurisdiction. 121,207 statutory citations are extracted from 50,666 Saudi judgments published in full text and matched to 15,855 articles across 290 instruments, giving the first article-level measurement of applied against enacted law for an Arabic-language legal system.

The applied law is a small and lopsided subset of the enacted one. 11.7 per cent of the statute book is ever cited — 30.5 per cent of the articles within the courts' own statutory jurisdiction — and 89.2 per cent of what is applied is procedural rather than substantive. The contrast is sharpest between two instruments that need no denominator: the law telling the commercial courts how to proceed has 99.0 per cent of its articles cited; the code telling them what the parties owe each other has 27.5 per cent.

Two results bear on why. Segmenting each judgment by its own headings, and attributing every citation to the court or to a party, locates the narrowing at the bench rather than the bar: the court's own reasons are more procedural than the arguments put to it, and substantive instruments are raised and answered with procedure. And amendment predicts neither citation nor its absence, so the unused portion of the book is not merely its neglected or superseded portion. Every measurement is generated by deposited code from a deposited corpus.

**Keywords:** empirical legal studies; legislation; judicial citation; legal informatics; open government data; Saudi Arabia

**JEL codes:** K40, K41, K10, C81

## 1 Introduction

A legislature enacts. A court applies. The two facts are studied by different literatures, using different data, and are rarely put side by side — not because the question is uninteresting but because answering it requires two corpora for the same jurisdiction, at the same level of granularity, and those seldom exist together. Legislation is studied where consolidated statutory text is machine-readable, chiefly the United States and Germany. Judicial behaviour is studied where opinions are digitised at scale, chiefly the same places. Where both exist, they are usually studied apart.

This article joins them for Saudi Arabia, a jurisdiction that has recently codified much of its private and commercial law and that publishes its commercial judgments in full text. On one side is a verified corpus of 15,855 articles across 290 instruments, each article carried with its number, its text and its legal status. On the other are 50,666 judgments, published by the Ministry of Justice, from which 121,207

statutory citations are extracted and matched to the article they cite. The result is a link between the enacted and the applied at the level at which a lawyer actually argues: not the statute, the article.

What that link shows is a legal system using a small and unrepresentative part of what it has enacted. 11.7 per cent of the articles in the corpus are cited at least once in fifty thousand judgments; 184 instruments are never cited at all. The obvious objection — that the published record is commercial and the denominator is not — is answered in §5.5 by rebuilding the denominator from the jurisdiction clause of the Commercial Courts Law itself, under two readings, and by a fourth denominator that concedes the objection entirely. The share moves; the conclusion does not.

The sharpest form of the result needs no denominator at all, because it compares two instruments to themselves. The Commercial Courts Law, which tells the court how to proceed, has 99.0 per cent of its articles cited. The Civil Transactions Law, the codification of Saudi private law and the instrument the jurisdiction clause names by paragraph, has 27.5 per cent of its 721. The procedural instrument is nearly exhausted; the substantive code is barely opened. Both are within the same court’s jurisdiction, both were enacted by the same legislature, and both are cited by the same judges in the same judgments.

Three further results follow. What is applied is overwhelmingly procedural (89.2 per cent of matched citations), and the most-cited provision in the entire corpus, cited 16,265 times, is the article stating which disputes the court may hear at all. The narrowing is not imposed by the parties: segmenting each judgment at its own headings shows the court’s reasons to be *more* procedural than the statement of the case, not less, and drawn from fewer instruments. And amendment activity predicts neither citation nor its absence, once the composition of the statute book is controlled for.

Four methodological points are reported here rather than left implicit, because each was a defect that produced confident wrong answers before it was found. The pattern that finds a citation must accept the prefixes Arabic attaches to the word for ‘article’; one that does not misses 15.7 per cent of everything, invisibly, because a citation never reached is not a citation failed on (§4.1). Matching a citation to an instrument is not string containment: a regulation’s title contains its parent law’s title, and a matcher that accepts containment in either direction silently misdirected roughly half of all citations in an earlier version of this analysis (§4.2). Anaphora — ‘the Law’, ‘the Regulation’, ‘that same Law’ — must be resolved by kind rather than by recency. And the segment of a judgment before the reasons is not the parties’ pleadings: it is written by the court, and at least 43.0 per cent of the statutory citations in it sit in a sentence whose actor is the court (§4.4). Each of these would change a reported result, and none is visible in the output of a matcher that reports only a match rate. The first was not found by any check run against this corpus at all; it was found by pointing the extractor at a document written outside it.

The article proceeds as follows. §2 places the work in three literatures. §3 describes both corpora and what the publishing interface does and does not provide. §4 sets out extraction, matching, article-number parsing and segmentation. §5 reports the results and the denominator robustness. §6 states what the data cannot support, and §7 asks what a legal system that uses a ninth of its own statute book is telling us about codification.

## 2 Related work

Three literatures meet here, and the gap this article occupies is between them rather than inside any one of them.

**Measuring legislation.** A quantitative literature treats a statute book as an object with size, structure and change over time. Katz and Bommarito (2014) measure the complexity of the United States Code through its structure, language and interdependence; Coupette et al. (2021) follow statutes and regulations in the United States and Germany over time as evolving networks. This work is careful about the enacted text and, necessarily, silent about use: nothing in a statutory corpus records whether anyone ever invoked a provision. The machine-readable legal corpora that have grown alongside it — Henderson et al. (2022), Niklaus et al. (2024) — are assembled to train language models and hold running text rather than the article-level status and identity a join of this kind requires.

**Measuring citation in judgments.** A second literature treats citation as evidence of what courts actually rely on. Fowler et al. (2007) build the citation network of United States Supreme Court precedent and use its structure to measure the legal importance of individual cases. The unit there is the precedent, and the network is a court citing itself. Statutory citation by courts has attracted much less attention, in part because it is less structurally interesting — a court citing a statute makes an edge to a different kind of object — and in part because in common-law systems the interesting citation is to cases. In a recently codified civil-law jurisdiction the reverse holds: what a court cites is overwhelmingly statute, and the question of which statute is not decorative.

**Selection: what reaches a court, and what is published.** Any claim made from decided cases has to confront two filters. Priest and Klein (1984) model the first: disputes that settle differ systematically from those tried, so litigated cases are not a sample of disputes. Siegelman and Donohue (1990) document the second by comparing published against unpublished employment discrimination cases and finding

them different in kind, not merely in number. Both filters apply here with full force, and neither can be removed by better extraction. They bound what the results mean rather than what they measure: this article measures what the published record contains, and §6 states plainly what that record is not.

**The Saudi legal system.** Scholarship on Saudi law has been overwhelmingly doctrinal and institutional, and for good reason: until recently there was nothing else to work with. Vogel (2000) remains the standard English-language treatment of the courts and their relation to Islamic legal doctrine, written before judgments were published at scale. The publication of commercial judgments at scale, and the codification programme of the last decade, have created materials that did not exist when that literature was written. This article is an attempt to use them quantitatively without pretending that the doctrinal questions have been answered by counting.

### 3 Data

#### 3.1 The legislative corpus

The legislative side is a verified corpus of Saudi primary legislation: 15,855 articles distributed over 290 instruments, each article carried as a record with its number, its heading, its text, and its legal status — original, amended, added or repealed. Instruments are laws (*nizam*) and their implementing regulations (*la’iha tanfidhiyya*), which in Saudi practice are separate instruments with independently numbered articles, a distinction that turns out to matter for matching (§4.2). Provenance for each record — the route by which it was retrieved, the corroboration behind it, and any discrepancy between sources — is carried in the corpus itself rather than asserted in prose.

#### 3.2 The judgment corpus

The judicial side is 50,666 judgments in full text, published by the Ministry of Justice and retrieved through the interface its own portal uses. Each record carries the judgment number, the court, the city, the Hijri date, and the text of the judgment as published, with the parties’ names already masked by the publisher.

Two features of the corpus bound everything that follows. It is overwhelmingly commercial: the commercial courts supply the great majority of the published judgments, first-instance and appellate, and other jurisdictions appear only in small numbers. And the interface exposes fields that would have made the analysis in §4.4 unnecessary — `judgmentFacts`, `judgmentReasons`, `judgmentRuling` —

which are null in every one of the 50,666 records. The structure the courts write into their judgments is present; the structure the publisher promises is not.

## 4 Method

### 4.1 Extracting a citation

A Saudi judgment cites a provision in a stable form: the article, then the preposition *min* ('of'), then the name of the instrument. Citations are extracted by matching that form and capturing two groups, the article expression and the instrument name. 121,207 citations are extracted from 41,395 of the 50,666 judgments (81.7 per cent); the remainder cite no statute in this form at all.

The capture needs a stopping rule. Judgments routinely quote the article they have just cited, and a name capture that runs to the end of the sentence swallows the quotation and matches nothing. Captures are therefore truncated at a list of tokens that begin a new syntactic unit — *al-sadir* ('issued by'), *wa-llati nassat* ('which provided'), *ala al-nahw al-ati* ('as follows') and others — before matching is attempted.

The pattern must also accept the prefixes Arabic attaches to the word for 'article' rather than setting apart: the same reference is written *al-maddah*, *bil-maddah*, *lil-maddah*, *wal-maddah*, each a single orthographic word. An earlier version of this analysis anchored on *al-maddah* alone, and so never saw *wafqan lil-maddah* ('pursuant to article') or *istinadan lil-maddah* ('relying on article') — among the commonest forms in judicial reasoning. The gap was worth 16,682 citations, 15.7 per cent of everything the analysis had then counted, and every figure in this article was recomputed after the correction.

How it was found is the part worth reporting. No check run against the judgments could see it: the match rate was unaffected, because a citation the pattern never reaches is not a citation it fails on, and the corpus shares the drafting conventions that made those forms rare within it. It surfaced when the same extractor was pointed at a document written by a practising lawyer, where it silently dropped two of his four articles. Fifty thousand documents produced by one institution share that institution's habits, and an instrument validated only against them inherits the habits as assumptions.

### 4.2 Matching an instrument

Instrument names in judgments differ from registry titles in predictable ways: the singular *nizam al-mahkama al-tijariyya* where the registry writes the plural *nizam al-mahakim al-tijariyya*; a regulation called *la'ihat nizam X* where the registry writes *al-la'iha al-tanfidhiyya li-nizam X*; and parenthetical variants in the registry itself. Each registry title is expanded into the forms a judgment might use, and a

captured name is matched against that index.

The direction of containment carries meaning, and getting it wrong produces confident errors rather than misses. A regulation's title contains its parent law's title as a substring, so a test that accepts containment in either direction will send a citation of article 16 of the Commercial Courts Law — the article stating the court's jurisdiction — to the Implementing Regulation of that same Law. In an earlier version of this analysis that single defect misdirected roughly half of all citations, and it was silent, because every citation still matched something. Matching therefore proceeds in three ordered steps: exact equality; then the longest registry title contained in the captured name, so that a citation spelling out the regulation's title reaches the regulation; then the shortest registry title containing the captured name, so that a citation naming only the law reaches the law.

Anaphora is resolved by kind. A bare *al-nizam* ('the Law') or *al-la'iha* ('the Regulation'), and phrases such as *dhat al-nizam* ('that same Law') or *al-la'iha nafsuha* ('the Regulation itself'), refer to an instrument named earlier in the same judgment. Resolving them to whichever instrument was named most recently sends articles of a law into its regulation whenever the regulation was mentioned last, which in a commercial judgment it usually was. *Al-nizam* is therefore resolved to the last *law* named in that judgment and *al-la'iha* to the last *regulation*. Anaphoric matches are recorded separately throughout: 108,474 citations name their instrument and 7,742 are resolved, and a resolved reference is a weaker observation that a reader should be able to set aside. 4.1 per cent of citations match no instrument in the registry and are dropped.

### 4.3 Reading an article number

Article numbers in Saudi judgments are more often spelled out than written in digits: *al-madda al-khamisa wa-al-tisun bad al-mi'a* is article 195. A parser that reads digits alone would measure a minority of citations and report it as a census. Article expressions are therefore parsed by a reader for Arabic ordinals that also handles Eastern Arabic numerals, compound forms, and both paragraph constructions — the compressed 16/3 and the expanded 'paragraph 3 of article 16' — returning an article number and, where present, a paragraph. 2.6 per cent of expressions cannot be parsed and are dropped. A further 1.2 per cent parse to a number beyond the last article of the matched instrument — a mismatch between the judgment and the registry, or a matching error — and are also dropped rather than assigned.

## 4.4 Separating the voices

A judgment is not one voice. It opens with a statement of the case, passes to the court's reasons, and closes with the operative disposition, and a statute cited in the first is not a statute the court applied. With the API's own fields empty, the segmentation is read from the headings the courts write: *al-waqai* (the statement of the case), then *al-asbab* (the reasons), then *hakamat al-daira* ('the circuit has adjudged'). Where the headings are present and in order the judgment is segmented; 27,321 judgments (53.9 per cent) qualify, and the rest are excluded from the voice analysis rather than forced into a shape they do not have. Because anaphora resolution needs document order, the text is scanned once and each citation is assigned to a segment by its offset.

One caution about the first segment matters enough to state as method. It is tempting to call it the pleadings, and wrong: *al-waqai* is written by the court, and it carries the court's own procedural narration alongside the parties' arguments as the court reports them.<sup>1</sup> Counting it as argument overstates the bar's citation practice and understates the bench's. Each citation in the segment therefore carries an attribution read from the sentence around it: whether the nearest actor to the citation is the court or a party. Because Arabic places the subject after the verb as readily as before it, the sentence is searched in both directions and the nearest cue wins.

The attribution cue is lexical, and its accuracy is reported rather than assumed. On a random sample of forty citations classified as the court's, 37 were the court narrating, two were ambiguous, and one was a party's memorandum written in the first person. The complementary class is much weaker, so no party share is reported anywhere in this article: what is reported is the court-attributed count, as a floor.

## 5 Results

### 5.1 How much of the statute book is applied

Of the 15,855 articles in the corpus, 1,849 are cited at least once in 50,666 judgments — 11.7 per cent of the statute book. At the level of the instrument, 106 of 290 instruments are cited at least once (36.6 per cent) and 184 are never cited at all.

The published record is heavily commercial, so a reader should expect criminal, family and administrative instruments to be absent. They are, and so is much else: entire commercial and financial instruments, with articles addressed to disputes that plainly arise, appear nowhere in fifty thousand

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<sup>1</sup>A typical instance, in translation: 'The circuit notes that it held this preparatory hearing pursuant to article 90 of the Implementing Regulation of the Commercial Courts Law' — the court describing a step it took, not an argument put to it.



Table 1: Statutory citations by segment, across 27,321 segmented judgments.

| Segment               | Citations | Instruments | Procedural |
|-----------------------|-----------|-------------|------------|
| Statement of the case | 22,185    | 80          | 84.9%      |
| Reasons               | 49,204    | 68          | 94.5%      |

judgments. Absence here is not proof that an instrument is unused; it is proof that its use, if any, is not visible in what the state publishes.

## 5.2 How concentrated the applied part is

The applied part is not merely small but concentrated inside itself. The single most-cited instrument accounts for 39.3 per cent of all matched citations, and the ten most-cited for 96.9 per cent. At article level the skew is sharper still: one article carries 14.6 per cent of all citations whose article number could be read, the top ten carry 43.3 per cent, and the top hundred 79.2 per cent.

What is applied is also overwhelmingly procedural. Instruments governing how a case is brought, heard, proved and enforced account for 89.2 per cent of matched citations; everything the law says about the substance of commercial obligations accounts for the rest.

The most-cited provision in the corpus, cited 16,265 times, is the article that lists the disputes the commercial court may hear at all. The provisions immediately behind it govern hearing a case in the absence of a defendant, the preparatory hearing, ordinary written evidence, and the plea of want of jurisdiction. The busiest provision in Saudi commercial adjudication is not a rule about commerce. It is the court's statement that it may proceed.

## 5.3 Who narrows: the bar or the bench

The obvious explanation for that narrowing is that courts answer what is put to them, and what is put to them is procedural. The segmentation tests it, and does not support it.

Table 1 shows the court's reasons to be more procedural than the statement of the case, not less, and drawn from a narrower set of instruments: 68 against 80. In 24 instruments a statute is cited in the statement of the case and never once in a court's reasons.

The attribution measurement sharpens this rather than softening it. Of the 22,185 citations in the statement of the case, at least 9,531 (43.0 per cent) sit in a sentence whose actor is the court itself. Those citations are 98.6 per cent procedural — the court reciting the provisions under which it convened, proceeded and satisfied itself of jurisdiction. The remainder, which is where the parties' arguments

live, is 74.6 per cent procedural. Leaving the court’s own narration inside the segment was flattering its procedural share; taking it out widens the gap between what is put to the court and what the court answers with, from 84.9 versus 94.5 to 74.6 versus 94.5.

The narrowing therefore happens at the bench, not at the bar. Substantive instruments are raised and are answered with procedure.

## **5.4 Amendment and application**

If legislative attention followed judicial use, the amended part of the statute book would be the applied part. It is not.

At instrument level, over the 266 instruments whose records carry a status breakdown — of which only 90 are cited at all — the rank correlation between an instrument’s churn and its citation count is  $\rho = .201$ : weakly positive and carried by a handful of large instruments.

At article level the pooled comparison appears to say the opposite of a null. Across 6,858 articles with a recorded status, 26.0 per cent of the 539 amended, added or repealed articles are ever cited, against 21.9 per cent of the 6,319 original ones, and an amended article draws 44.3 citations on average against 12.9 — a ratio of  $3.4\times$ .

That ratio is a composition effect and should not be reported without its control. Amended articles are not spread evenly across the statute book, and the instrument carrying the largest share of all citations is also among the most heavily amended; the pooled figure mostly records which instruments are litigated, not which articles are amended. Controlling within the instrument — one vote per instrument, over the 46 instruments holding both amended and original articles — amended articles are more likely to have been cited in 20 instruments, less likely in 25, with 1 tie: a two-sided sign test gives  $p = .551$ . Inside an instrument, whether an article has been amended tells you nothing about whether a court has ever applied it.

## **5.5 Choosing the denominator**

A share is only as good as what it is divided by, and the objection to the first result is obvious: the published judgments are overwhelmingly commercial, so a denominator holding the whole registry counts instruments a commercial court could never reach. Dividing by them makes the applied law small by construction.

The objection is answered by measurement, and the scope is drawn by the statute rather than by the author. Article 16 of the Commercial Courts Law enumerates what the court may hear, and its paragraphs

Table 2: Article-level coverage under four denominators. The article-level join covers the instruments whose article records are on disk, so four instruments cited at instrument level but with no article number that parses into range do not appear here.

| Denominator              | Instruments | Articles | Ever cited |
|--------------------------|-------------|----------|------------|
| The whole registry       | 290         | 15,855   | 11.7%      |
| Article 16 scope, broad  | 115         | 8,356    | 20.6%      |
| Article 16 scope, narrow | 54          | 5,297    | 30.5%      |
| Instruments ever cited   | 102         | 9,049    | 20.4%      |

name their instruments: partnership contracts under the Civil Transactions Law (§3), the Companies Law (§4), the Bankruptcy Law (§5), the intellectual property laws (§6), and ‘the other commercial laws’ (§7). Only the last requires judgement, so it is drawn twice — narrowly, as the instruments that constitute trade itself, and broadly, as everything whose subject is trade, finance or the regulation of commercial activity — and both readings are reported. Table 2 adds a fourth denominator that concedes the objection entirely: the instruments courts are observed to cite at all, which asks how much of an instrument is used even where the court does go.

The finding survives every denominator that can honestly be constructed. On the narrowest reading of the court’s own jurisdiction — 5,297 articles across 54 instruments, every one of them within Article 16 — 1,615 articles are ever cited: 30.5 per cent. Restricting to the instruments courts actually do cite leaves 20.4 per cent. The share moves with the denominator, as it must; what does not move is that most of the law the court is given goes unused.

Inside instruments the same gradient appears again, and it is sharper than anything at the level of the instrument. The Commercial Courts Law has 95 of its 96 articles cited, 99.0 per cent; then 90.7 per cent of the Evidence Law; 71.2 per cent of the Companies Law; 51.9 per cent of the Bankruptcy Law. And 27.5 per cent of the Civil Transactions Law — 198 of 721 articles — the codification of Saudi private law, and the instrument §3 of Article 16 points to by name.

That contrast is the result stated as sharply as the data allow. The instrument that tells the court how to proceed is almost exhausted; the instrument that tells it what the parties owe each other is barely opened.

## 6 Threats to validity

**The corpus is what the Ministry publishes, and that is not the docket.** Two selection filters stand between any dispute and this dataset, and neither can be removed by better extraction. Disputes that settle never reach a judgment at all, and settled disputes differ systematically from tried ones (Priest and Klein 1984). Of the judgments that are given, publication is a further filter, and published and unpublished

decisions have been shown to differ in kind rather than merely in number (Siegelman and Donohue 1990). Every result here is a statement about the published record. Where the article says a provision is never cited, it means never cited in what the state publishes.

**The record is commercial, urban and recent.** 95.0 per cent of the judgments come from the commercial courts; 85.1 per cent come from three cities; the Hijri dates run from 1422 to 1448, but 87.9 per cent of the corpus is dated 1442 or later. Criminal, family, labour and administrative adjudication are effectively absent — the corpus holds 14 judgments of a labour court and no decisions of the tax and zakat committees. An instrument addressed to those forums cannot be cited here, which is why §5.5 rebuilds the denominator rather than defending the first one.

**The registry is a corpus, not the statute book.** The 290 instruments are the instruments this project has verified, not the totality of Saudi legislation. Ministerial decisions, circulars of the Supreme Judicial Council and unpublished administrative instruments are outside it, and some of them are cited in judgments: they account for part of the 4.1 per cent of citations that match nothing. The denominators in Table 2 are therefore denominators of a verified corpus.

**Published fields cannot be relied on, including one that looks reliable.** The interface exposes `judgmentFacts`, `judgmentReasons` and `judgmentRuling`, and all three are null in every record, which is why the segmentation in §4.4 is read from the text. It also exposes `isAppeal`, which is present, well-typed, and false on every one of the 50,666 records — including all 2,296 judgments whose own court field names a court of appeal or the Supreme Court. A field that never varies is not a field. First-instance and appellate judgments are therefore separated here by court name and not by the flag provided for the purpose.

**The portal serves some judgments more than once.** 1,303 records (2.6 per cent) are exact textual duplicates of another record, across 796 groups, leaving 49,363 distinct texts. Duplication inflates any count of citations that repeats a standard formula, which is precisely the material the busiest articles consist of. The counts reported here are counts of citations in the corpus as published, and the shares were recomputed over distinct texts alone to check that the concentration is not an artefact of the repetition: the most-cited instrument's share moves from 39.3 per cent to 39.4, the top ten from 96.9 to 96.9, and the procedural share from 89.2 to 89.1. A reader comparing an absolute count against another source should still know the duplication is there.

**Matching, parsing and resolution each drop or infer.** 4.1 per cent of citations match no instrument; 2.6 per cent carry an article expression that cannot be parsed; 1.2 per cent parse to a number the matched instrument does not have and are dropped rather than assigned. A further 7,742 of the 121,207 citations name no instrument at all and are resolved by inference from the same judgment; they are counted separately throughout so that a reader who rejects the inference can subtract them. Neither loss is uniform, and the direction matters more than the size. The unparsed rate is flat over time — between 1.1 and 3.1 per cent in every Hijri year from 1439 onwards — so nothing is being lost from one period in particular. An unmatched instrument name, by construction, is more likely for an instrument outside the registry, and an instrument outside the registry is not one of the heavily cited ones; that loss falls on the long tail, which makes the concentration reported here a conservative estimate rather than an inflated one.

**The attribution inside the statement of the case is a floor.** The court-versus-party classification rests on lexical cues, validated at 37 of 40 on a hand-read sample of the court class. The complementary class was measured and found unreliable, so no party share is reported. Where the article says at least 43.0 per cent of those citations are the court’s own narration, the qualifier is doing real work.

## 7 Discussion

The result that survives every robustness check is a difference between two instruments of the same legal system, enacted by the same legislature, addressed to the same court, and cited by the same judges in the same judgments. One is nearly exhausted. The other is barely opened.

The tempting reading is that the substantive code is defective, or unknown to the profession, or too new. The data do not support any of the three, and one of them can be tested directly: if the code were simply unknown to the bar, citations to it would be scarce in the parties’ material as well as in the court’s reasons. They are not. Substantive instruments are raised in the statement of the case at a markedly higher rate than they appear in the reasons, and 24 instruments appear there and never once in a court’s reasoning. Whatever is narrowing the applied law is operating between the argument and the judgment.

A second reading is more mundane and probably closer: a judgment must resolve a dispute, and a judgment that resolves it on jurisdiction, admissibility, evidence or limitation never reaches the substantive question at all. On that reading the procedural concentration is not a pathology but the arithmetic of disposition — most cases are decided on the shortest available ground, and the shortest available ground is procedural. This is consistent with the finding that the most-cited provision in the corpus is the jurisdiction clause, and with the preparatory-hearing article being cited overwhelmingly in narration rather than in

reasoning.

Both readings have the same implication for how codification is evaluated. A legislature that measures its output by articles enacted is measuring something only loosely related to what its courts use. Between them, the enacted and the applied here run in parallel: at instrument level the rank correlation between churn and citation is weak and carried by a few large instruments ( $\rho = .201$ ), and inside an instrument, where composition cannot do the work, amendment carries no signal at all. If a code is intended to shape adjudication, the fact that 27.5 per cent of its articles have ever been cited in fifty thousand judgments is a measurement worth having, whatever one concludes from it.

The method generalises beyond this jurisdiction more readily than the substance. Any state that publishes judgments in full text and holds its legislation at article level can be measured this way, and the defects documented in §4 — containment direction, anaphora by kind, the recital that is not the parties' voice — are not features of Arabic. They are features of the way legal texts refer to other legal texts, and each of them, left uncorrected, produces a confident number rather than a missing one.

## 8 Conclusion

Saudi Arabia has enacted a great deal of law in a short time and has begun to publish what its commercial courts do with it. Joining the two shows a legal system applying a small, procedurally concentrated part of what it has enacted: 11.7 per cent of the articles in a verified corpus of 15,855, 30.5 per cent within the courts' own statutory jurisdiction, and 89.2 per cent of all matched citations falling on the instruments that govern how a case is heard rather than what it is about. The narrowing is not imposed by the parties, and it does not track legislative attention.

The sharpest statement of it needs no denominator at all. The law that tells the court how to proceed has 99.0 per cent of its articles cited. The code that tells the court what the parties owe each other has 27.5 per cent.

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## Use of AI tools

The analysis code deposited with this article, and drafts of this manuscript, were produced with the assistance of a large language model (Anthropic’s Claude) working under the author’s direction. The author designed the study, framed every question, verified each result against the underlying corpus, and is responsible for the content. No measurement in this article is produced by a language model: every number is computed by the deposited code from the deposited data, and the manuscript is typeset from a generated file of macros so that a figure in the prose cannot drift from the figure in the data. Three claims that an earlier draft made were withdrawn during this checking, and one segmentation defect was found and corrected the same way.

## Availability of data and code

This article is written to the data and code availability policy of the American Economic Association. The legislative corpus, the judgment corpus, the extraction and matching code, the ordinal parser, the segmentation and attribution code, and the scripts that produce every number in this article are public.<sup>2</sup>

The judgments are reproduced as the Ministry publishes them, with parties’ names masked at source;

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<sup>2</sup>Repository: <https://github.com/Abdullah-m7/saudi-legal-corpus-ai>. Every measurement reported here is generated into a single file of macros by `make_numbers.py` and typeset from it; no number in this manuscript is typed by hand.

a further redaction pass over identifiers that the publisher's own masking missed is included in the repository and documented there.

## **Declarations**

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**Competing interests.** None.

**Author contributions.** Sole author.